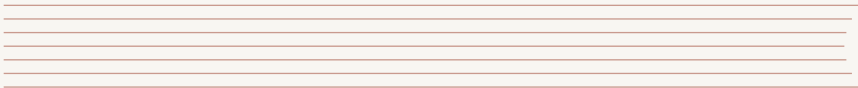


# DATA WAREHOUSE

Air Quality, Economy and Health in Europe (2010–2020)

Francesco Innocenzi 2087246 Luca Festugato 2064594  
Sapienza University of Rome

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midcentury  
modern



# OUTLINE

- Introduction and Dataset
- Conceptual Design (DFM)
- Starflake Logical Schema
- Reconciled Layer: 3-Tier ELT
- OLAP Analysis and Insights
- Conclusions

## ■ The starting idea

Is there an observable relationship between **air pollution**, **economic wealth**, and **mortality** across European countries? Answering this requires a schema able to connect different domains on the same geographic and time basis — this is why we chose a Data Warehouse.

Environment (Air Quality), Economy (GDP), and Health (Mortality) are usually analyzed separately: combining them into a single multidimensional model enables comparisons that would otherwise be impossible — for example, understanding whether a richer country also breathes cleaner air and lives longer. Ten-year coverage across Europe (2010–2020).

## DOMAINS AND DATA SOURCES

| Area               | Source   | Files                                      | Format |
|--------------------|----------|--------------------------------------------|--------|
| Air Quality        | WHO      | 1 (city, PM2.5/PM10/NO2)                   | Excel  |
| Mortality (HFAMDB) | WHO      | 4 ( <i>All causes</i> + 3 specific causes) | CSV    |
| Economy (GDP)      | Eurostat | 1                                          | TSV    |
| Population         | Eurostat | 1                                          | TSV    |

Even data from the same source come in different formats (GDP and Population are 2 separate Eurostat files, not one): structures, country codes, and granularity do not match — a cleaning and integration step is needed before loading them into the same Data Warehouse.

## A CLOSER LOOK AT THE SOURCE FIELDS

### **WHO — Air Quality (Excel) (2000–2021)**

IS03, WHO Country Name, City or Locality, Measurement Year, PM2.5, PM10, NO2, % time coverage — one row per (city, year).

### **Eurostat — GDP, estat\_nama\_10\_pc.tsv (1975–2025)**

freq, unit, na\_item, geo, then one year per column: *wide* format.

### **Eurostat — Population, estat\_demo\_pjan.tsv (1960–2025)**

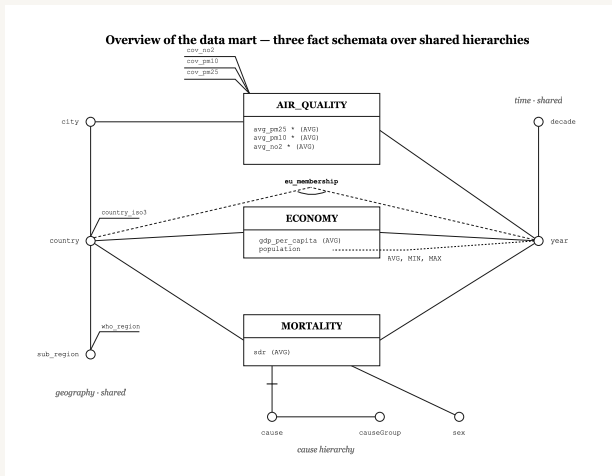
freq, unit, age, sex, geo, then one year per column: a separate file from GDP, with different columns — they are only joined at the reconciled stage.

### **WHO — HFAMDB Mortality (CSV, 4 files) (1968–2022)**

COUNTRY, COUNTRY\_GRP, AGE\_GRP\_LIST, SEX, SUBNATIONAL\_MDB, YEAR, VALUE: one row per (country, sex, year), one file per cause of death.

# THE CONCEPTUAL MODEL (DFM)

- **Air Quality:** granularity City / Year. Measures include PM2.5, PM10, and NO2 levels, along with their respective temporal coverage.
- **Economy:** granularity Country / Year. Measures include GDP per capita and population.
- **Mortality:** granularity Country / Year / Gender / Cause. The measure is the Standardized Death Rate (SDR), it is aggregated with AVG, never with SUM. CAUSE is an optional dimension: populated only by the specific diseases.



# THE CONSTELLATION MODEL

## ■ Why it is a *constellation* and not 3 separate schemas

TIME and GEOGRAPHY are **conformed dimensions**: shared by the 3 facts with the same identifiers and hierarchies. This is what enables **Drill-Across** queries – comparing avg\_pm25, gdp\_per\_capita, and sdr by country and year, even though they come from different facts.

eu\_membership is a **cross-dimensional** attribute: it depends on the combination (Country, Year), not on a single dimension – EU membership changes over time, as shown by two real transitions in the 2010–2020 window.

| Country | Year        | eu_membership |
|---------|-------------|---------------|
| HRV     | 2012 → 2013 | Non-EU → EU   |
| GBR     | 2019 → 2020 | EU → Non-EU   |

## GRANULARITY AND BUS MATRIX

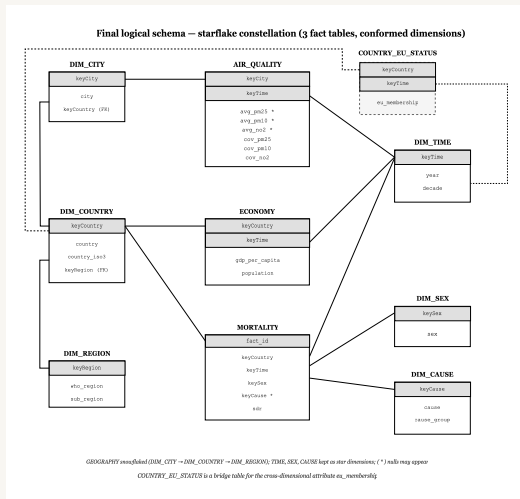
| Fact        | GEOGRAPHY | TIME | SEX | CAUSE |
|-------------|-----------|------|-----|-------|
| AIR_QUALITY | City      | Year | -   | -     |
| ECONOMY     | Country   | Year | -   | -     |
| MORTALITY   | Country   | Year | Sex | Cause |

### ■ Grain per fact and conformed dimensions

- Different grain per fact: AIR\_QUALITY at **City** level (urban measurements), ECONOMY/MORTALITY at **Country** level (national data)
- Shared conformed dimensions: GEOGRAPHY and TIME (year → decade) enable cross-domain analysis

## Implementation Details

- **Constellation structure** for the three fact tables.
- **Snowflake for Geography:**  
The geographic hierarchy is normalized into three separate tables linked by foreign keys.
- **Star for the other dimensions:**  
Time, Sex, and Cause are kept "flat" and denormalized in single tables.
- **Surrogate** keys assigned to each dimension table.



## Why Snowflake for Geography

city → country → sub\_region: avoids repeating the country name/ISO3 for hundreds of cities (redundancy), and handles the two different grain levels (city for air quality, country for economy/mortality) — a single flat DIM\_GEOGRAPHY could not do this cleanly.

## Why Star for Time, Sex, Cause

These are **small, static** tables (few rows, almost never updated): normalizing them like geography would add extra JOINS with no real benefit, since there is no significant redundancy to remove. Denormalization here is a pure win for reads.

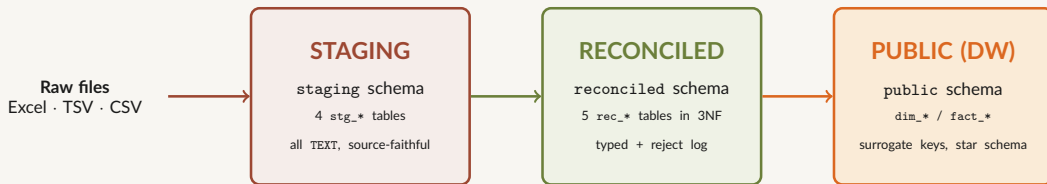
## ■ The problem

Only **3 specific diseases** belong to DIM\_CAUSE. The total entry "*All causes*" is not a fourth cause: it is the overall total.

## ■ Solution: optional key

- "*All causes*" → `key_cause = NULL` in the MORTALITY fact
- Integrity guaranteed in PostgreSQL by **two partial unique indexes**:  
`uq_mortality_cause` and `uq_mortality_total`
- They prevent double counting between totals and per-cause values

## 3-TIER ELT ARCHITECTURE



We adopt a **3-tier ELT**: raw data is loaded immediately, as-is, into the staging schema; cleaning and integration happen *afterwards*, with declarative SQL queries inside PostgreSQL, moving through the reconciled and public tiers.

## TIER 1: STAGING

### Python's role changes: only *ingestion*, never *cleaning*

Python reads Excel/TSV/CSV files and writes them *exactly as they are* into the staging schema: no parsing, no casting. Everything stays TEXT on purpose — a NUMERIC cast would immediately fail on Eurostat flags ("31200 p", ":"); casting becomes a decision postponed to the next step.

| Domain      | Staging table        | Source                     |
|-------------|----------------------|----------------------------|
| Air quality | stg_who_aap          | 1 WHO Excel file           |
| GDP         | stg_eurostat_gdp     | 1 Eurostat TSV file        |
| Population  | stg_eurostat_pop     | 1 Eurostat TSV file        |
| Mortality   | stg_hfamdb_mortality | 4 HFAMDB CSV files (UNION) |

### ■ From *staging* to *reconciled*: cleaning and integration

- **Cleaning:** non-interpretable values are discarded; *wide* format (one year per column) → *long* format (one row per year)
- **Integration:** GDP and Population (2 separate Eurostat files) are merged into a single row per country-year, without losing data when one source does not cover a given country/year
- Tables in **3NF**: **natural** keys (e.g. `country_iso3`), constraints (PK, FK, CHECK), and specific types instead of the TEXT used in staging

| reconciled.* table | Grain (1 row per ...)       |
|--------------------|-----------------------------|
| rec_country        | Country                     |
| rec_city           | (Country, City)             |
| rec_air_quality    | (Country, City, Year)       |
| rec_economy        | (Country, Year)             |
| rec_mortality      | (Country, Year, Sex, Cause) |

## Every discarded row leaves a trace

Every rejection creates an explicit row in `reconciled.rec_reject_log`: table, key, reason, and timestamp — so we can reconstruct what was excluded and why, without re-running the pipeline.

| Reason for rejection                | Example         |
|-------------------------------------|-----------------|
| Eurostat aggregate, not a country   | EU27_2020, EA20 |
| ISO3 code outside geographic Europe | AFG, USA        |

## `v_reject_summary`: the aggregated log

A view that groups the log by table and reason, counting occurrences: one query answers "how many rows were excluded, and why".

### ■ From reconciled to public: surrogate key lookup

The `reconciled.*` tables feed the dimensions and facts, replacing natural keys (ISO3, city, year) with surrogate keys. At the end of the ELT process, the `public` schema contains the fully populated fact constellation, ready for analytical queries.

| From reconciled.*            | To public.*                                    |
|------------------------------|------------------------------------------------|
| <code>rec_country</code>     | <code>dim_region, dim_country</code>           |
| <code>rec_city</code>        | <code>dim_city</code>                          |
| <code>rec_time</code>        | <code>dim_time</code>                          |
| <code>rec_air_quality</code> | <code>air_quality (fact)</code>                |
| <code>rec_economy</code>     | <code>economy (fact), country_eu_status</code> |
| <code>rec_mortality</code>   | <code>mortality (fact)</code>                  |

For mortality, the *"All causes"* row does not match any specific cause: `key_cause` automatically stays NULL.

## ■ Inconsistent country codes

WHO/HFAMDB use **ISO3** (3 letters), Eurostat uses **geo** with **2 letters** (2 exceptions: EL = Greece, UK = United Kingdom): a dedicated mapping table is needed to bring them all back to one common key.

- `sub_region` (Northern/Southern/Eastern/Western Europe): does not exist in the original datasets, defined by hand to enable geographic roll-up
- `cause_group` (e.g. *cardiopulmonary*, *cancer*): the 3 specific causes of death were grouped from scratch for a more natural drill-down
- Cleaning numbers: Eurostat flags ("31200 p" = provisional, ":" = missing) are cleaned before casting to numeric

# ROLL-UP

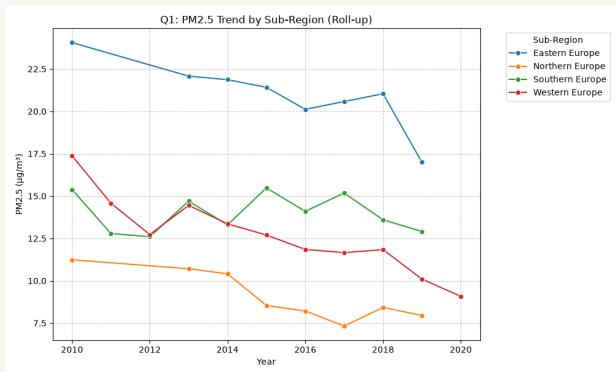
```
SELECT r.sub_region,
       t.year,
       COUNT(DISTINCT ci.key_city)      AS n_cities,
       ROUND(AVG(aq.avg_pm25), 2)      AS pm25,
       ROUND(AVG(aq.avg_pm10), 2)      AS pm10,
       ROUND(AVG(aq.avg_no2), 2)      AS no2
FROM   air_quality aq
JOIN   dim_city    ci ON ci.key_city    = aq.key_city
JOIN   dim_country co ON co.key_country = ci.key_country
JOIN   dim_region  r  ON r.key_region   = co.key_region
JOIN   dim_time    t  ON t.key_time     = aq.key_time
GROUP BY r.sub_region, t.year
ORDER BY r.sub_region, t.year;
```

**Roll-up** query that aggregates air quality data. Air quality is stored at city grain. We climb the geographic hierarchy city → country → sub\_region and aggregate to the region level, by year.

# ROLL-UP INSIGHT: PM2.5 TREND

## Roll-up

Trend of PM2.5 levels over time across different sub-regions in Europe.



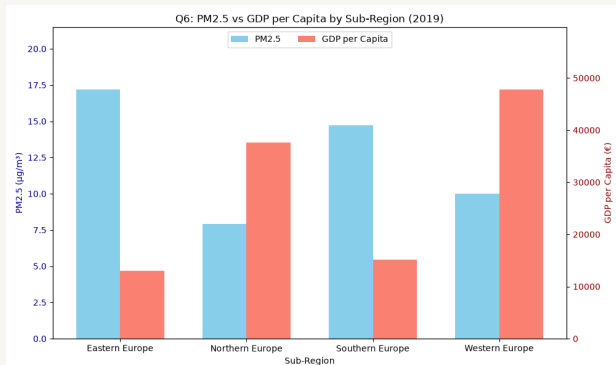
# DRILL-ACROSS

```
WITH aq_country AS (
    SELECT ci.key_country, aq.key_time,
           AVG(aq.avg_pm25) AS pm25, AVG(aq.avg_no2) AS no2, COUNT(*) AS n_cities
    FROM   air_quality aq
    JOIN   dim_city ci ON ci.key_city = aq.key_city
    GROUP BY ci.key_country, aq.key_time
),
resp AS (
    SELECT m.key_country, m.key_time, m.sdr
    FROM   mortality m
    JOIN   dim_cause ca ON ca.key_cause = m.key_cause
    JOIN   dim_sex s ON s.key_sex = m.key_sex
    WHERE  ca.cause = 'Respiratory diseases' AND s.sex = 'ALL'
)
SELECT r.sub_region, t.year,
       COUNT(DISTINCT a.key_country) AS n_countries,
       SUM(a.n_cities) AS n_cities,
       ROUND(AVG(a.pm25)::numeric, 2) AS pm25,
       ROUND(AVG(re.sdr)::numeric, 2) AS respiratory_sdr,
       ROUND(AVG(e.gdp_per_capita)::numeric, 0) AS gdp_per_capita
FROM   aq_country a
JOIN   dim_country co ON co.key_country = a.key_country
JOIN   dim_region r ON r.key_region = co.key_region
JOIN   dim_time t ON t.key_time = a.key_time
LEFT JOIN resp re ON re.key_country = a.key_country AND re.key_time = a.key_time
LEFT JOIN economy e ON e.key_country = a.key_country AND e.key_time = a.key_time
GROUP BY r.sub_region, t.year
HAVING COUNT(DISTINCT a.key_country) >= 3
ORDER BY r.sub_region, t.year;
```

# POLLUTION AND WEALTH: THE CONSTELLATION AT WORK

## Drill-Across

Joining AIR\_QUALITY and ECONOMY on the conformed dimensions GEOGRAPHY and TIME: in 2019 Eastern Europe has the highest PM2.5 ( $17.2 \mu\text{g}/\text{m}^3$ ) and the lowest GDP per capita (€12,970) – Western Europe shows the exact opposite ( $10.0 \mu\text{g}/\text{m}^3$ , €47,767).



## ■ In summary

- An effective **ROLAP Constellation** architecture for connecting heterogeneous domains (air, economy, health)
- DFM design → Starflake logical schema → 3-tier ELT pipeline (staging → reconciled → public), reproducible and auditable
- Advanced OLAP analyses reveal the link between pollution, GDP, and mortality

## Moving to the live demo